

Hostel Room Allocation and Maintenance Register

SIH 2026 - Internal Practical Assessment

1. The Problem

- **Current State:** Hostel uses printed lists for room allocation and logs complaints verbally at the desk.
- **Consequences:**
 - Nobody knows which rooms are vacant quickly.
 - Complaints get lost, and outstanding issues are forgotten.
 - Students complain repeatedly because past logs aren't visible.
 - Management has no oversight over repair times.

2. The Solution

A unified, digital register that tracks real-time room occupancy and securely logs maintenance complaints with full history, ensuring no issue is ever lost and management can immediately identify urgent outstanding faults.

3. Application Screenshots

(Insert screenshot of the Add Room and Log Complaint Form here)

(Insert screenshot of the Complaint Register Listing here)

Note: The application features a glassmorphic dark theme tailored for clarity and ease of use, prioritizing urgent outstanding complaints at the top of the list.

4. How Derived Figures Are Calculated

- **Days Outstanding:** Calculated dynamically on the server by taking the difference between the current timestamp (`julianday('now')`) and the `created_at` timestamp of the complaint.
- **Hours Outstanding:** Derived by multiplying the days outstanding by 24.
- **Logic:** Server-side calculation ensures all users see exactly the same figure regardless of their local timezone or browser settings.

5. Project Status

- **What Works:**
 - End-to-end room registration and complaint logging.
 - Server-side validation and derived figure calculation.
 - Listing ordered by outstanding status and age.
 - Status history tracking in a separate database table.
 - Docker containerization.
- **What is Unfinished:**
 - Authentication for desk clerks/wardens.
 - Demonstration video recording.

6. Future Improvement

- **Improvement:** Implement automated email/SMS notifications to the maintenance staff when a complaint becomes "urgent" (e.g., > 48 hours outstanding). This would proactively reduce the time students wait for critical repairs.